

Submission CALC1-PS2-SUB06

Question CALC1-PS2-Q01

notation changes halfway:

1. $f'(x) \approx 2x(6x-1)+(x^2+4)6$
2. $f'(2) \approx 4(11)+(8)6 \approx 22$

marks, subscripts, and units are mixed up

result written as: 22 (please interpret)

Question CALC1-PS2-Q02

notation changes halfway:

1. $\int 2x \, dx \approx (2/2)x^2$
2. $[2/2 x^2]_0^2 \approx 4$

marks, subscripts, and units are mixed up

result written as: 4 (please interpret)

Question CALC1-PS2-Q03

notation changes halfway:

1. $2w+2l \approx 20 \Rightarrow l \approx 10-w$
2. $A(w) \approx w(10-w)$; at $w \approx 7$, $A \approx 21$
3. A maximum requires $A'(w) \approx 0$, so the stated dimensions are not assumed optimal without that check.

marks, subscripts, and units are mixed up

result written as: $A(w)=w(10-w)$; $A(7)=21$; check $A'(w)=0$ (please interpret)

Question CALC1-PS2-Q04

notation changes halfway:

1. $x^2-9 \approx (x-3)(x+3)$
2. cancel $x-a$ only after identifying $x \neq a$
3. $\lim_{x \rightarrow 3} x+3 \approx 6$

marks, subscripts, and units are mixed up

result written as: 6 (please interpret)

Question CALC1-PS2-Q05

notation changes halfway:

1. Critical point: $f'(c) \approx 0$ or $f'(c)$ is undefined (when f is defined).
2. A sign change in f' from $+$ to $-$ indicates a local maximum.

marks, subscripts, and units are mixed up

result written as: critical point: $f'(c)=0$ or undefined; $+$ to $-$ sign change gives a local maximum (please interpret)

Question CALC1-PS2-Q06

notation changes halfway:

1. $f'(x) \approx 2x(1x-1)+(x^2+3)1$
2. $f'(3) \approx 6(2)+(12)1 \approx 19$

marks, subscripts, and units are mixed up
result written as: 19 (please interpret)

Question CALC1-PS2-Q07

notation changes halfway:

1. $\int 4x \, dx \approx (4/2)x^2$

2. $[4/2 x^2]_0^5 \approx 50$

marks, subscripts, and units are mixed up
result written as: 50 (please interpret)

Question CALC1-PS2-Q08

notation changes halfway:

1. $2w+2l \approx 16 \Rightarrow l \approx 8-w$

2. $A(w) \approx w(8-w)$; at $w \approx 5$, $A \approx 15$

3. A maximum requires $A'(w) \approx 0$, so the stated dimensions are not assumed optimal without that check.

marks, subscripts, and units are mixed up

result written as: $A(w)=w(8-w)$; $A(5)=15$; check $A'(w)=0$ (please interpret)